

CATHOLIC JUNIOR COLLEGE
2008 JC2 PRELIMINARY EXAMINATION

Name: _____ Class: _____ Calculator Model No: _____

Question	1	2	3	4	5	6	7	8	9	10	11	12	Total
Marks													

MATHEMATICS

Paper 1

8863/01

10 September 2008 (Wednesday)

1300 – 1600 hr (3 hours)

Additional Materials: Answer Paper
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise. Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **4** printed pages.

Section A: Pure Mathematics [40 marks]

1. Given that the point $P(2, 4)$ lies on the curve $y = x^3 - 3x + 2$, find the equation of the normal at P , giving your answer in the form $ax + by = c$, where a , b and c are integers. [3]

2. The functions f and g are defined as follows:

$$f: x \rightarrow x^2 + 2x, \quad x \geq -1$$

$$g: x \rightarrow x + 4, \quad x \in \mathbb{R}$$

- (i) Show that the composite function gf exists. [2]
 (ii) Find the value of x such that $gf(x) = 7$ [3]
 (iii) Find, in similar form, the inverse function f^{-1} . [3]

3. (a) Solve the following simultaneous equations:

$$\begin{aligned} \log_2(6x + 3y) &= 2 + \log_2 6 \\ 3^{x+1} &= 27(3^{y-1}) \end{aligned} \quad [5]$$

- (b) On the same axes, sketch the graphs of $y = \cos x$ and $y = 2 \cos 3x$ for $0 \leq x \leq \pi$, labeling each graph carefully and showing the coordinates of the intersections of the graphs with the axes.

Hence solve the inequality $\cos 3x - \frac{1}{2} \cos x > 0$ for $0 \leq x \leq \pi$. [4]

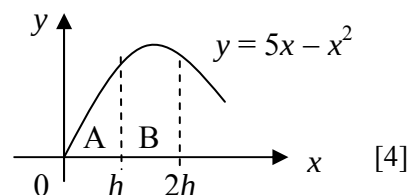
4. (a) Find, simplifying your answers,

(i) $\int (2 - 3x)^5 dx$ [2]

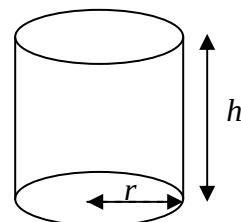
(ii) $\int 6e^{\frac{x}{2}} dx$ [2]

(iii) $\int \frac{2}{2-x} dx$ [2]

- (b) The region A is bounded by the curve $y = 5x - x^2$, the x -axis and the line $x = h$. The region B is bounded by the curve $y = 5x - x^2$, the x -axis, and the lines $x = h$ and $x = 2h$ as shown in the diagram. Given that the area of B is twice the area of A, find the value of h .



5. In order to minimise the use of plastic and save the environment, a company redesigned a 500 cm^3 container for its drinks. The new design is a closed cylindrical container with height h cm and radius r cm as shown.



- (i) Show that the exterior surface area, $S \text{ cm}^2$, of the container is given by $S = 2\pi r^2 + \frac{1000}{r}$. [3]
 (ii) Given that r can vary, find the stationary value of S to two decimal places, and determine the nature of this stationary value. [4]
 (iii) Assume that as the container is being manufactured, S and r vary with time. Given that r is increasing at the rate of 3 cm s^{-1} , find the rate at which S is increasing when $r = 6 \text{ cm}$. [3]

Section B: Statistics [60 marks]

6. The mass of one random bar of chocolate of Brand X follows a normal distribution with mean μ grams and standard deviation σ grams. Given that 1 out of 10 bars of chocolate, on average, weigh more than 85 grams and 1 out of 5 bars of chocolate, on average, weigh less than 75 grams, find μ and σ , correct to 2 decimal places. [5]
7. A school wants to engage a new canteen vendor and would like to know the food preferences of the student population. The school has 850 girls and 900 boys, and a sample of 75 is needed.
- Explain clearly how a stratified sample can be obtained. [3]
 - State one advantage of using a stratified sample compared to a random sample in this case. [1]
 - What modification is needed to your method in part (i) if a quota sample is to be obtained? [1]
8. Under usual conditions, the absorption time of a commonly prescribed drug is said to have mean of 5.35 hours. When the drug is administered to a random sample of 1000 adults, the absorption time, x hours, of each adult is measured. The results are summarized by $\Sigma x = 5384$ and $\Sigma x^2 = 29210$.
- Find unbiased estimates of the population mean and variance, correct to 3 decimal places. [2]
 - Assuming a normal distribution, test at the 5% significance level whether the population mean absorption time is greater than 5.35 hours. [4]
 - State, giving a reason, whether it is necessary to assume a normal distribution for the test to be valid. [1]
9. When a machine is used to dig up potatoes, there is a probability of 0.1 that a potato will be damaged in the process.
- Write down an expression for the probability that a random selection of 12 potatoes dug up by the machine will include 3 damaged potatoes and find its value. [2]
 - Find the least value of n such that the probability of obtaining at least 3 damaged potatoes in a random sample of n potatoes exceeds 0.06. [3]
 - By using a suitable approximation, find the probability of getting at least 3 damaged potatoes out of a random sample of 100 potatoes. [3]
10. A student conducted a study on the population growth of a city in the 1990s with data sourced from a statistical website. The table below shows the data obtained.

No. of years, x (after 1990)	1	2	3	4	5	6	7	8	9
Population, y (in thousands)	3135	3231	3314	3419	3525	2561	3796	3927	3959

- Give a sketch of the scatter diagram for the data, as shown on your calculator. Giving a reason, identify a data pair which should be regarded as suspect. [2]
- Replace the suspect data pair with (6, 3671), and calculate the product moment correlation coefficient for the revised data. Comment on your answer [2]
- Using the revised scatter diagram, comment on the trend observed in the population growth in the 1990s. [1]
- Find both the lines of regression of y on x , and x on y for the revised data. [2]
- The student would like to make an estimate for the population size in the year 2008. Using the appropriate line of regression, calculate this estimate. Comment briefly on the reliability of this estimate. [3]

11. (a) The events A, B and C are such that A and B are independent, and A and C are mutually exclusive. If $P(A) = 0.4$, $P(B) = 0.2$, $P(C) = 0.3$ and $P(B \cap C) = 0.1$, calculate:
- (i) $P(A \cap B)$ [1]
 - (ii) $P(A \cup B)$ [1]
 - (iii) $P(C | B')$ [2]
- (b) Jane travels to school by bus on dry mornings and takes the taxi on wet mornings. The probability of a wet morning occurring is $\frac{3}{8}$. The probability of Jane being late if she takes the bus is $\frac{1}{6}$. The probability of Jane being late if she takes the taxi is $\frac{1}{24}$.
- (i) Find the probability that Jane is late on a random day. [2]
 - (ii) Find the probability that it is a wet morning given that Jane is late. [2]
 - (iii) Taking a week as having 5 schooling days, find the probability that Jane is late for 2 days in a random week. [3]

12. Japanese-imported Rockmelons and Watermelons are sold by weight. The weights, in grams, of Rockmelons and Watermelons are modeled as having independent normal distributions with means and standard deviations as shown in the table.

	Mean weight	Standard deviation
Rockmelon	800	50
Watermelon	1600	180

- (i) Two melons, one of each type, are chosen at random. Find the probability that the Rockmelon weighs between 700 grams and 900 grams, and the Watermelon weighs between 1500 grams and 1700 grams. [3]
- (ii) Find the probability that the total weight of 4 randomly chosen Rockmelons is more than twice the weight of a randomly chosen Watermelon by at least 850 grams. [4]
- (iii) A Rockmelon with weight exceeding m grams is considered as “Premium”. Calculate the least value of m such that the probability of a randomly chosen Rockmelon being “Premium” is less than 0.05. Give your answer to the nearest gram. [3]

Rockmelons are sold at \$0.05 per gram and Watermelons at \$0.03 per gram.

- (iv) Find the probability that the total selling price of a randomly chosen Rockmelon and a randomly chosen Watermelon is more than \$75. [4]

End of Paper